

ITDS12063 – SOFTWARE ENGINEERING

1st Year,2nd Semester, 2025

Assignment Title:

Sri Lankan Wild Animals Guide- System Design Report

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1. INTRODUCTION

1.1 Problem Identification

Sri Lanka is renowned as a biodiversity hotspot with unique endemic species and diverse ecosystems. However, public awareness about wildlife conservation and the island's natural heritage remains limited. There is a need for an accessible, informative platform that educates visitors about Sri Lankan wildlife, national parks, and conservation efforts while providing an engaging user experience.

1.2 Purpose and Objectives

The primary purpose of this website is to create an educational platform that showcases Sri Lanka's rich biodiversity and promotes wildlife conservation awareness. The specific objectives are:

- To provide comprehensive information about Sri Lankan mammals, birds, and their habitats
- To showcase national parks and wildlife sanctuaries across the island
- To create an interactive gallery featuring wildlife photography
- To facilitate user engagement through contact forms and feedback mechanisms
- To promote environmental conservation and ecotourism

1.3 Target Audience

The website targets multiple user groups:

- **Nature Enthusiasts:** Individuals interested in wildlife and biodiversity
- **Tourists:** Local and international visitors planning wildlife tours
- **Students and Researchers:** Academic users seeking information about Sri Lankan fauna
- **Conservationists:** Environmental activists and conservation organizations
- **General Public:** Anyone interested in learning about Sri Lanka's natural heritage

2. TECHNOLOGY STACK

2.1 Frontend Technologies

Technology	Version/Details	Purpose
HTML5	Latest standard	Structure and semantic markup
CSS3	Custom stylesheet	Styling and responsive design
JavaScript	ES6+	Client-side interactivity and dynamic features

2.2 Backend Technologies

Technology	Version/Details	Purpose
PHP	7.4+	Server-side processing and database operations
MySQL	8.0+	Database management system
Apache/XAMPP	Latest	Local development server

2.3 Development Tools

- **Code Editor:** Visual Studio Code / Sublime Text
- **Browser Testing:** Chrome, Firefox, Edge
- **Database Management:** phpMyAdmin
- **Diagram Draw:** Draw.io

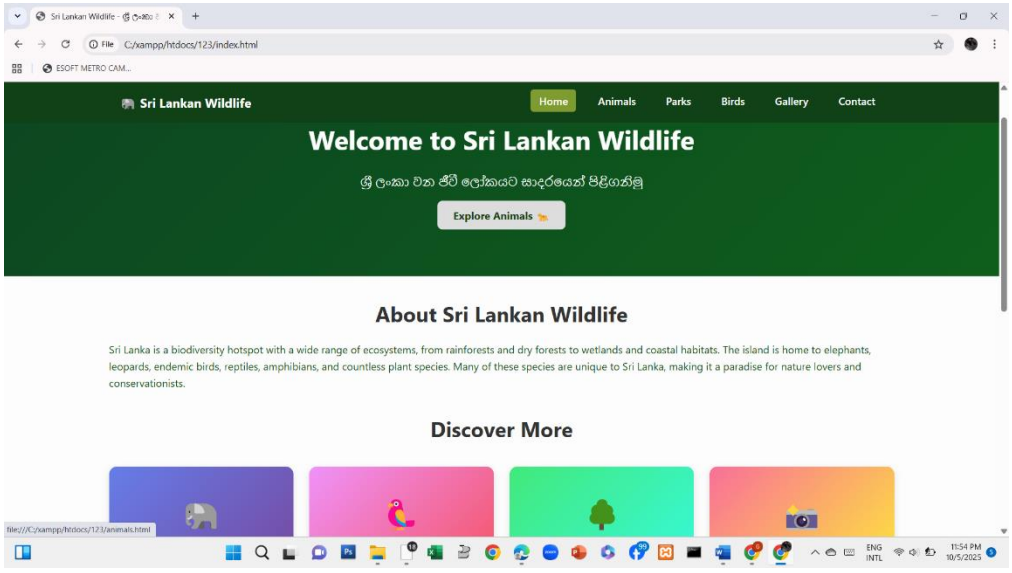
3. WEBSITE STRUCTURE AND PAGES

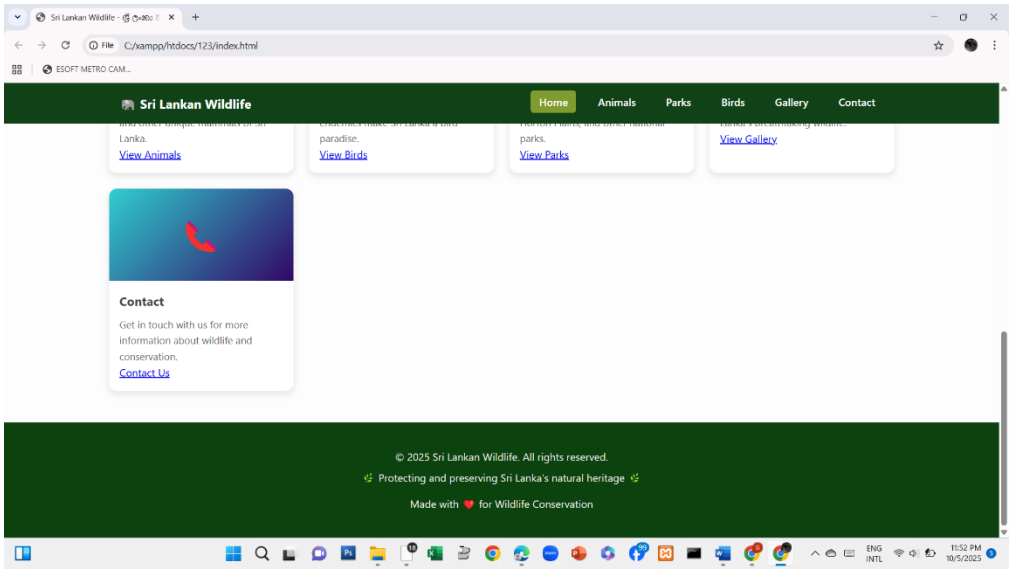
3.1 Page Overview

The website consists of 6 main pages, each serving a specific purpose:

3.1.1 Home Page (index.html)

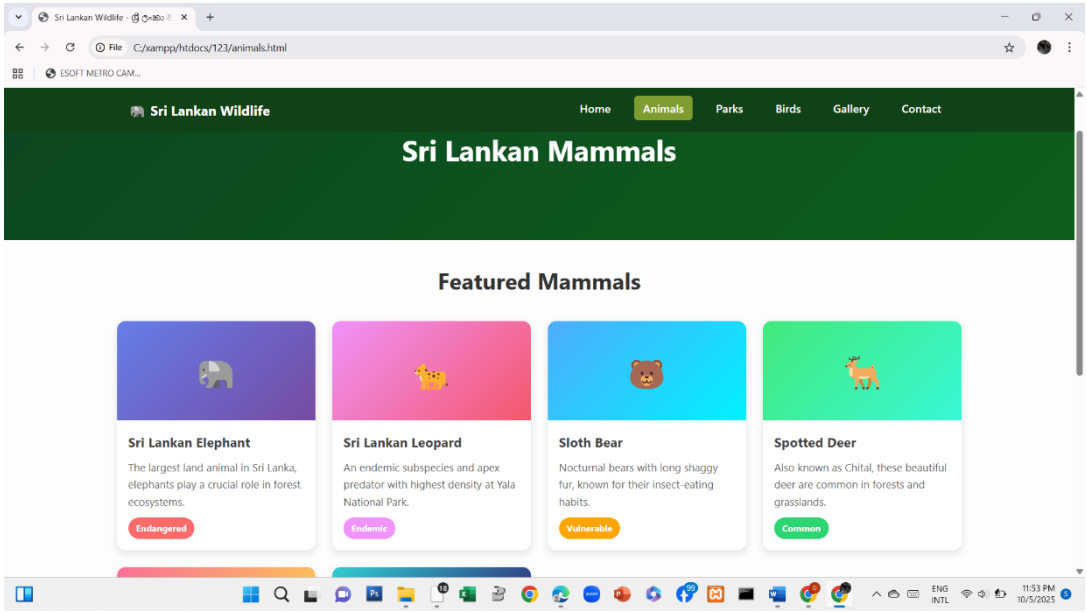
The landing page welcomes visitors with a hero section featuring Sri Lankan wildlife imagery. It provides an overview of the website's content and quick navigation links to all major sections. The page includes cards linking to Animals, Birds, Parks, Gallery, and Contact pages.

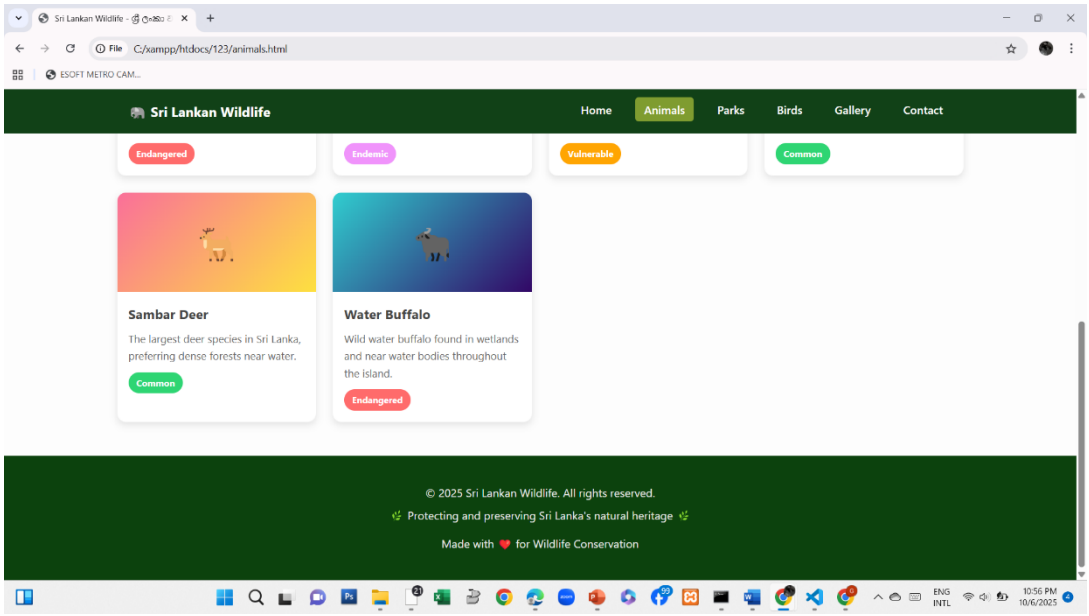




3.1.2 Animals Page (animals.html)

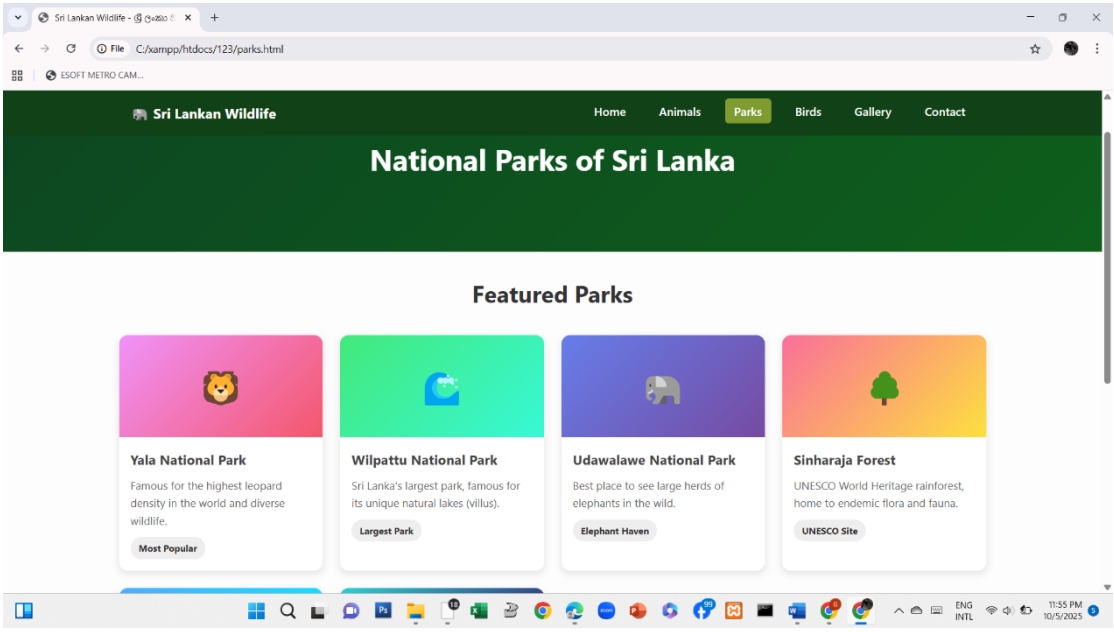
This page showcases Sri Lankan mammals including elephants, leopards, sloth bears, spotted deer, sambar deer, and water buffalo. Each animal card displays key information and conservation status. Clicking on an animal opens a modal with detailed facts, scientific names, and habitat information.

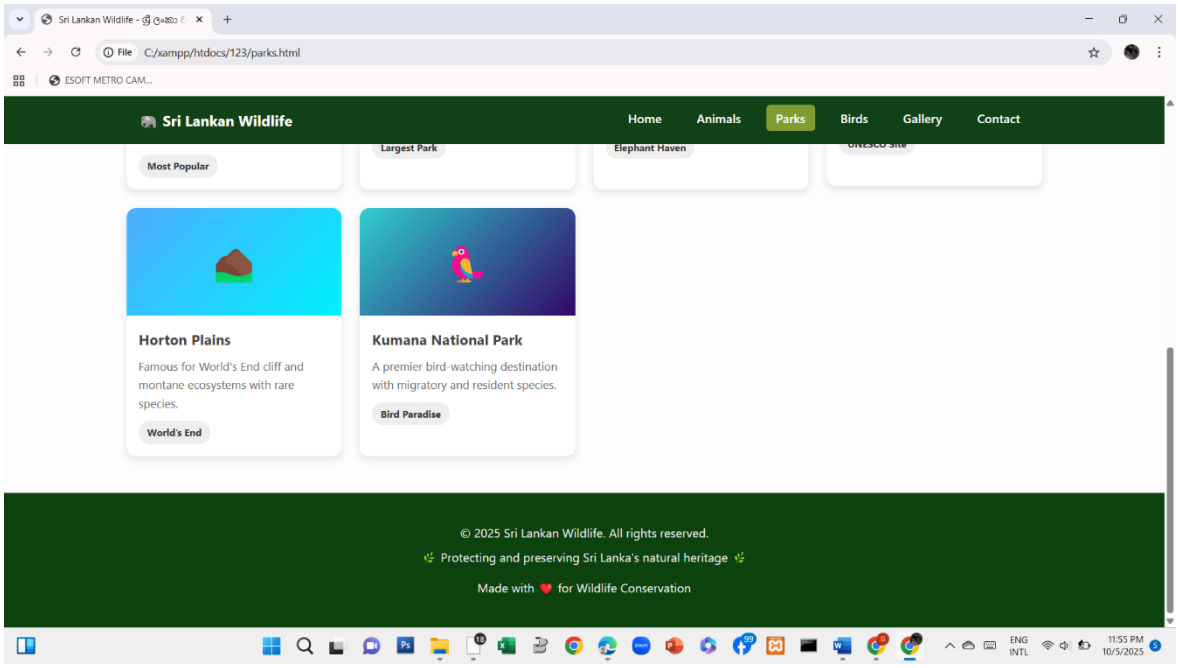




3.1.3 National Parks Page (parks.html)

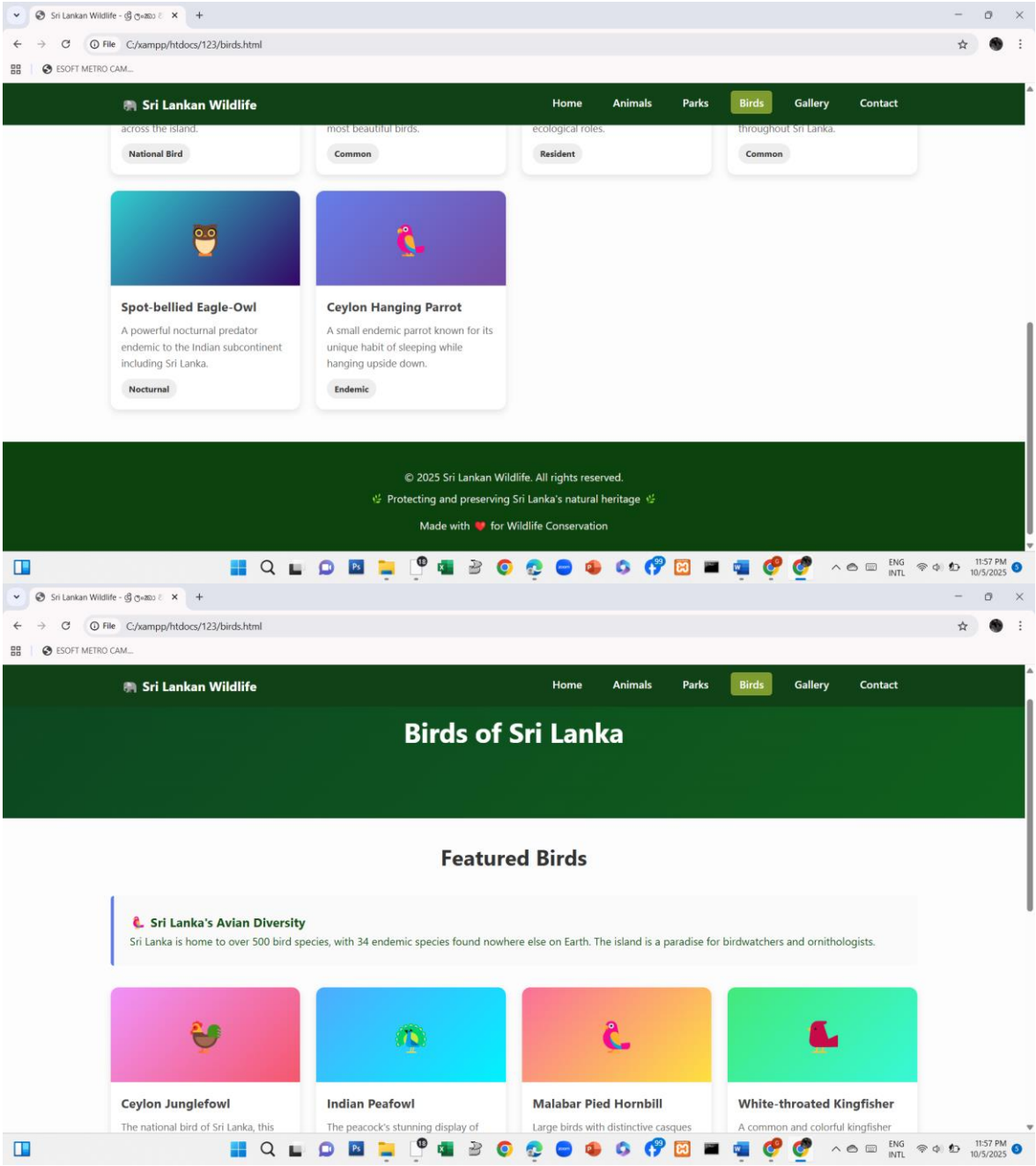
Features six major national parks: Yala, Wilpattu, Udawalawe, Sinharaja, Horton Plains, and Kumana. Each park card includes area information, unique features, and best visiting times. External links direct users to official park websites.





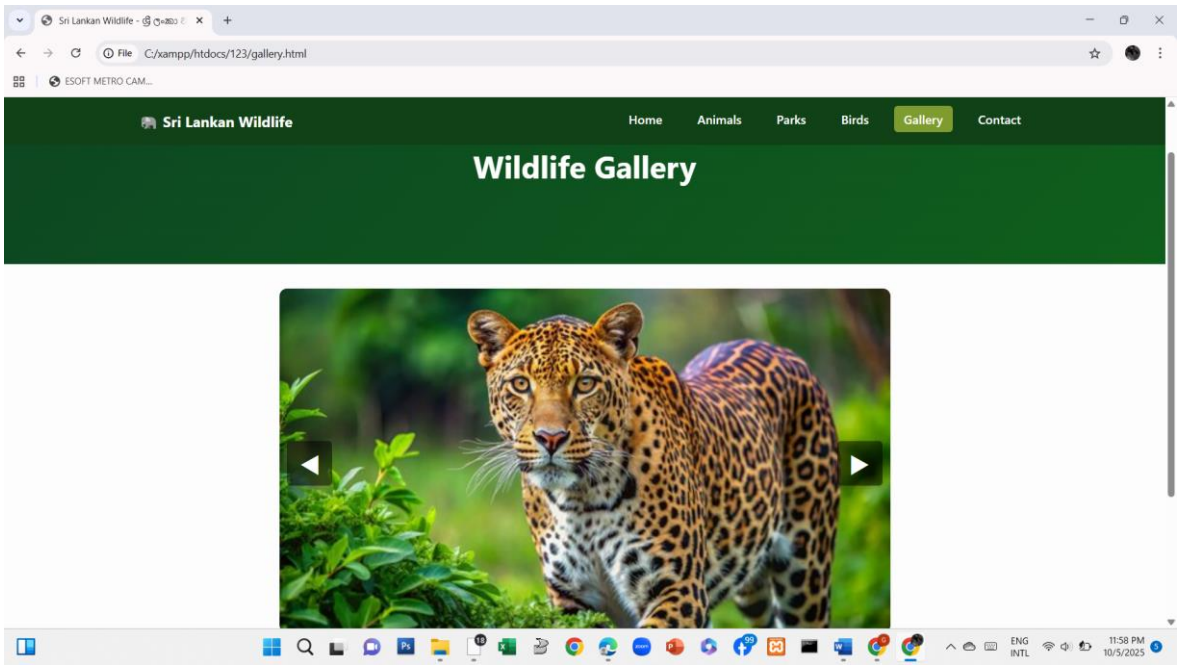
3.1.4 Birds Page (birds.html)

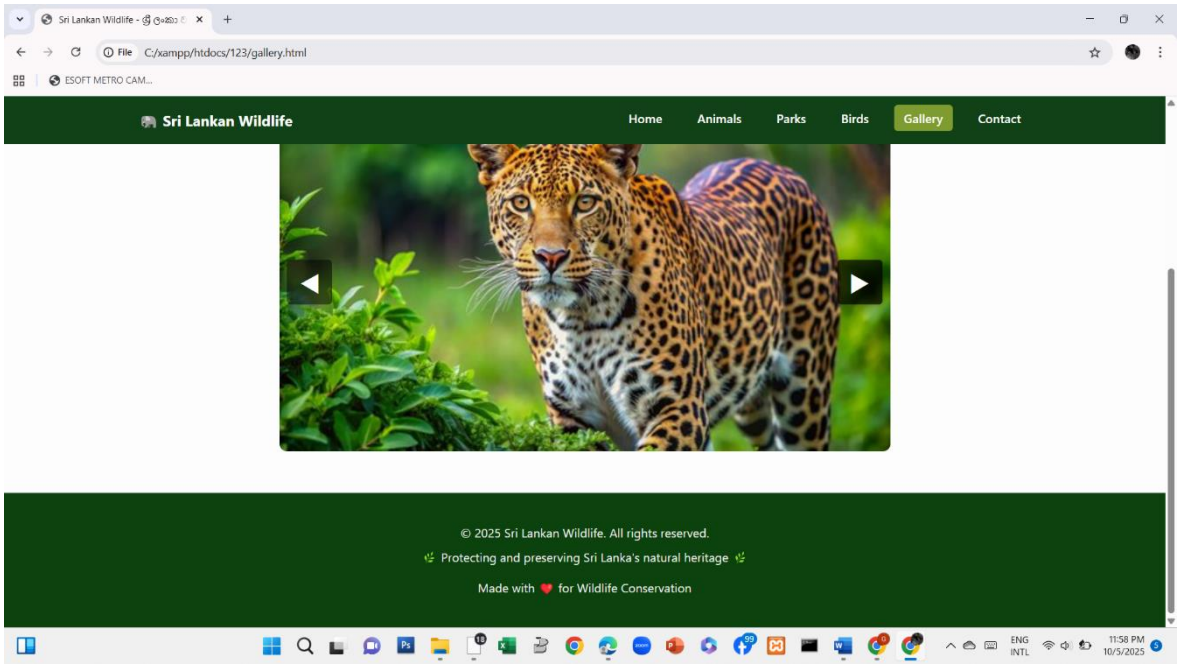
Presents Sri Lanka's avian diversity with featured species including the Ceylon Junglefowl (national bird), peacock, hornbill, kingfisher, owl, and hanging parrot. Interactive modals provide detailed bird information.



3.1.5 Gallery Page (gallery.html)

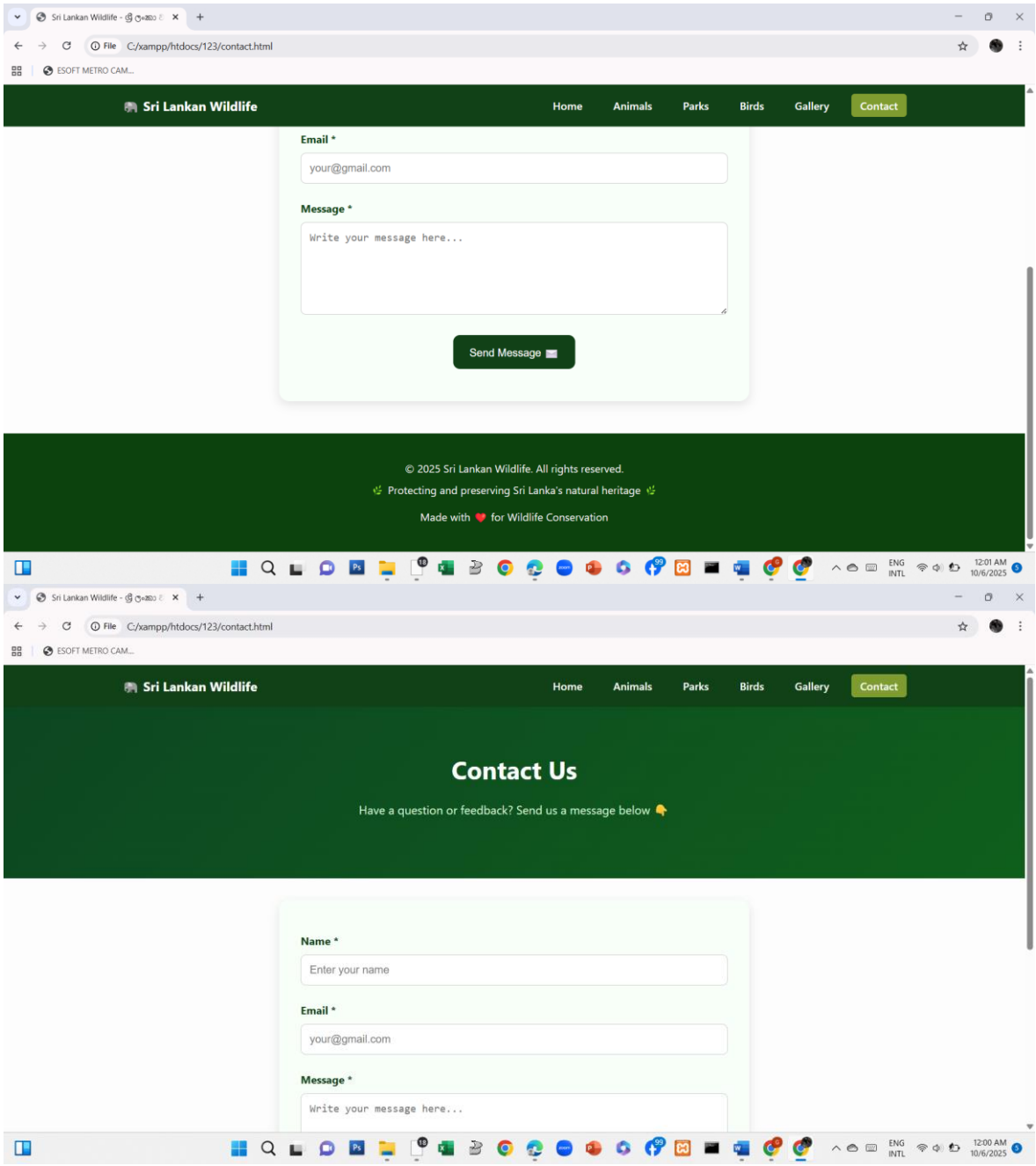
An interactive image slider showcasing wildlife photography from across Sri Lanka. Features automatic transitions and manual navigation controls for browsing through 19+ high-quality wildlife images.



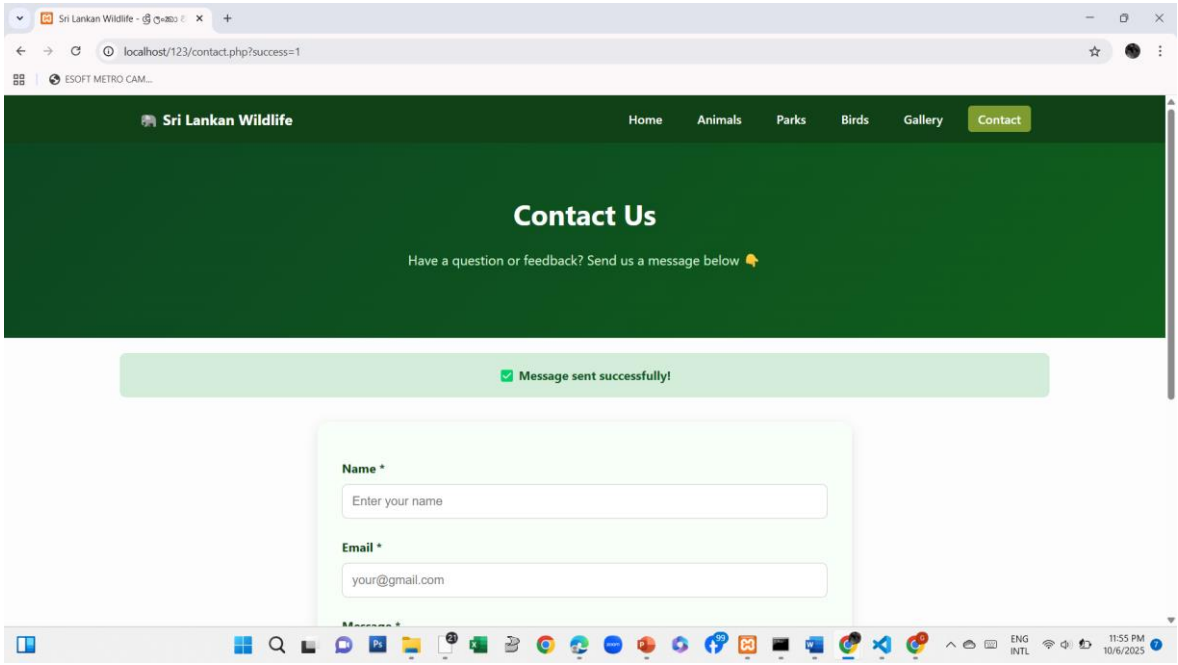


3.1.6 Contact Page (contact.html)

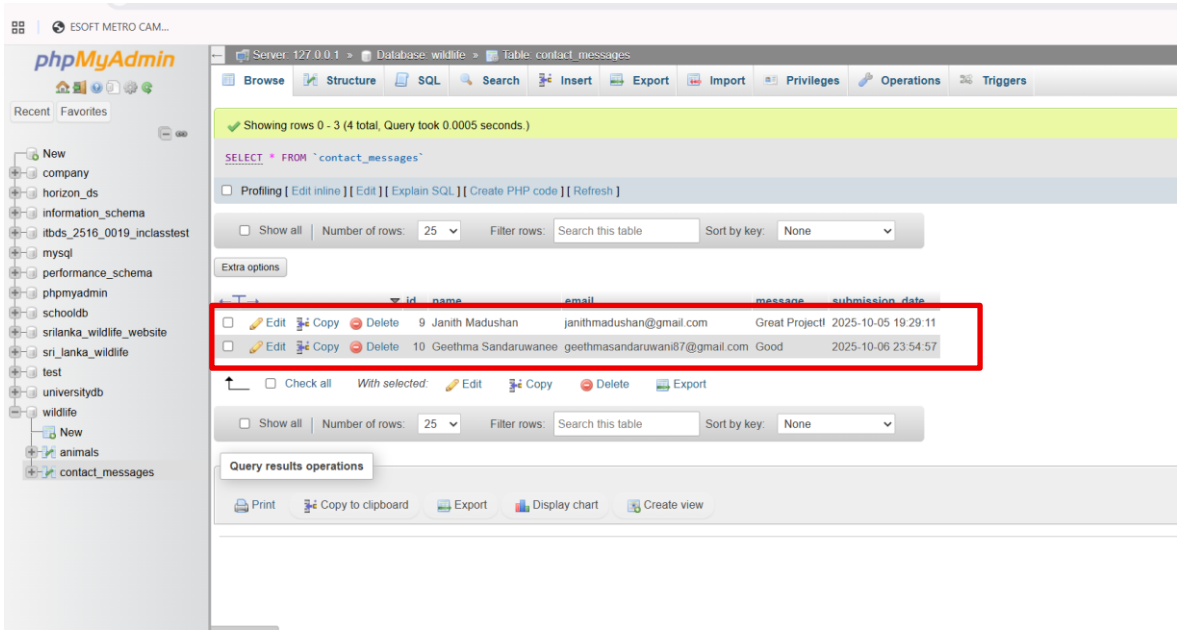
A form-based page allowing visitors to send messages, feedback, or inquiries. Includes client-side validation and backend PHP processing with database storage. Displays success/error messages after form submission.



The message that appears after filling out the contact form says "Successfully".



The details of those who fill out the contact form are shown in a red box.



4. KEY FEATURES IMPLEMENTED

4.1 JavaScript Features

4.1.1 Image Slider (Gallery Page)

Implementation: An automatic image carousel with manual navigation controls. Images transition every 4 seconds with smooth fade effects. Users can navigate using previous/next buttons.

Code Logic: Uses an array of image paths with index-based navigation. The setInterval() function enables automatic progression, while modulo arithmetic ensures circular navigation.

4.1.2 Modal Popups (Animals, Birds, Parks Pages)

Implementation: Dynamic modal windows display detailed information when users click on animal, bird, or park cards. Each modal is populated with specific data from JavaScript objects.

Code Logic: Event listeners trigger showDetails() functions that inject HTML content into the modal container and toggle CSS classes for visibility.

4.1.3 Form Validation (Contact Page)

Implementation: Client-side validation ensures all required fields are filled before submission. HTML5 validation attributes enforce email format and required fields.

Code Logic: Uses HTML5 required attributes and type="email" for basic validation, with server-side PHP validation as backup.

4.1.4 Interactive Navigation

Implementation: Active page highlighting in the navigation menu shows users their current location. Consistent navigation across all pages.

Code Logic: CSS class "active" is applied to the current page's navigation link, styled with distinct colors.

4.1.5 Dynamic Content Display

Implementation: Animal, bird, and park data is stored in JavaScript objects and dynamically rendered into modal popups.

Code Logic: Data structures contain all information, and template literals generate HTML on demand.

4.1.6 Click Outside to Close Modal

Implementation: Users can close modal windows by clicking outside the content area or using the close button.

Code Logic: Window onclick event listener detects clicks on the modal backdrop and triggers the closeModal() function.

4.2 Backend Integration

4.2.1 Contact Form Processing

Implementation: PHP script (contact.php) processes form submissions, validates input, and stores messages in the MySQL database.

Database Table: contact_messages (id, name, email, message, created_at)

Security Features: Prepared statements prevent SQL injection, server-side validation ensures data integrity.

4.2.2 Database Connection

Implementation: MySQLi connection established at the beginning of contact.php with error handling.

Configuration: Localhost server, 'wildlife' database, root user credentials.

4.2.3 Feedback Messages

Implementation: URL parameters display success, error, or validation messages after form submission using PHP conditional statements.

5. DATABASE DESIGN

5.1 Database Schema

Database Name: **wildlife**

Table 1: contact_messages

Field Name	Data Type	Constraints	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each message
name	VARCHAR(255)	NOT NULL	Sender's name
email	VARCHAR(255)	NOT NULL	Sender's email address
message	TEXT	NOT NULL	Message content
Submission_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Submission timestamp

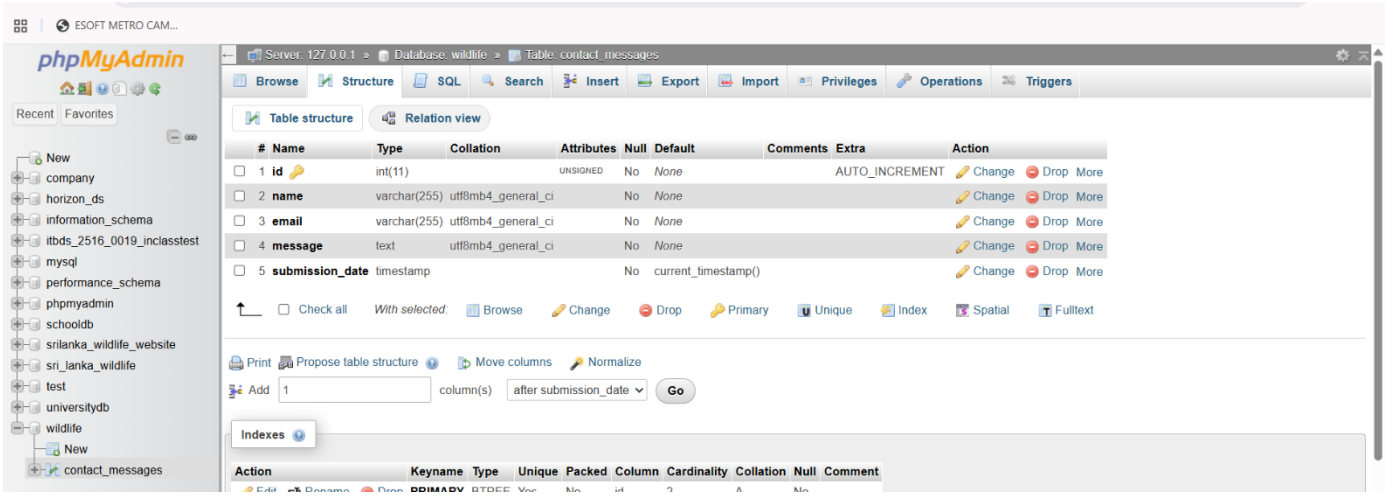
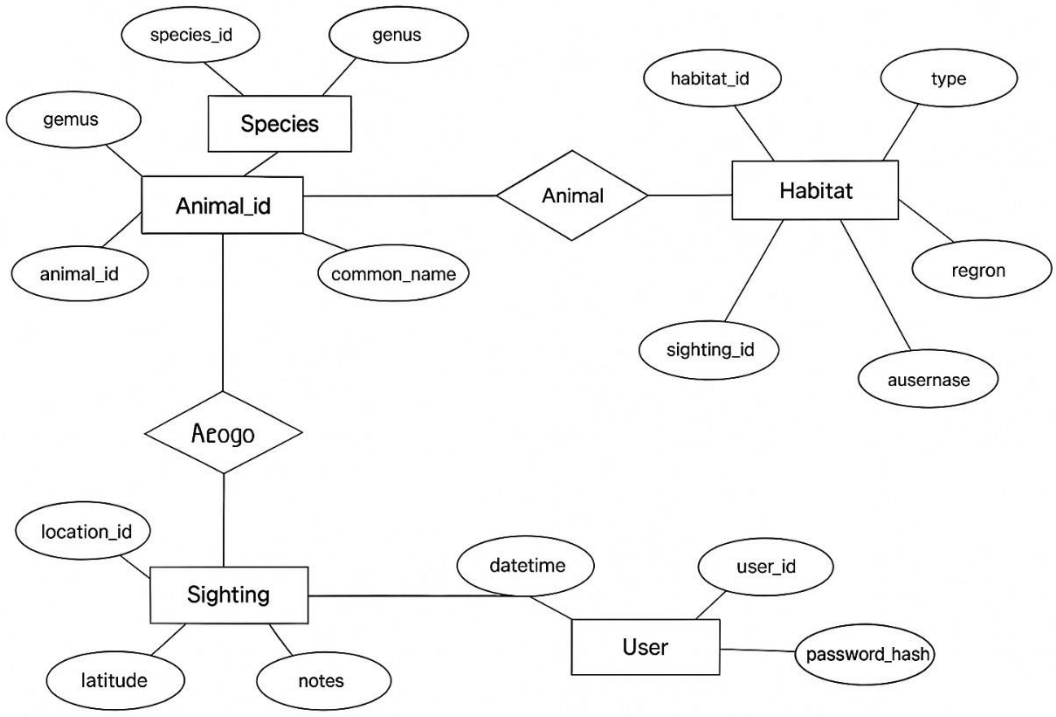


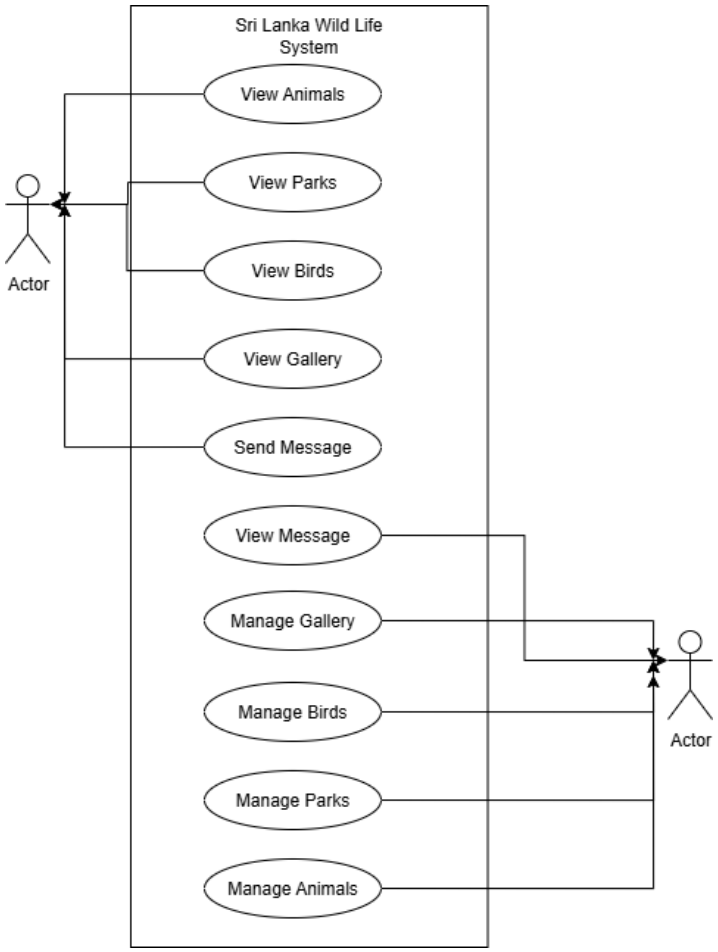
Table 2:

Field Name	Data Type	Constraints / Attributes	Description
id	INT(10)	UNSIGNED, NOT NULL, AUTO_INCREMENT, PRIMARY KEY	A unique identifier for each animal entry. Using AUTO_INCREMENT means the database automatically assigns the next number.
name	VARCHAR (255)	NOT NULL	The common name of the animal (e.g., "Asian Elephant"). This is mandatory.
scientific_name	VARCHAR (255)	NULL	The binomial nomenclature (e.g., "Elephas maximus"). Allowed to be empty (NULL).
description	TEXT	NULL	A longer field for details about the animal, its behavior, etc. TEXT can store more characters than VARCHAR.
conservation_status	ENUM(...)	NOT NULL	The specific status of the species' risk of extinction. The ENUM type restricts values to a predefined list (e.g., 'Endangered', 'Vulnerable', 'Common'). This is mandatory.
habitat	VARCHAR (255)	NULL	A brief description of the animal's natural environment (e.g., "Tropical Rainforests").
image_url	VARCHAR (255)	NULL	The URL/path to an image of the animal.
created_at	DATETIME	NULL, DEFAULT CURRENT_TIMESTAMP	The date and time the record was added to the database. CURRENT_TIMESTAMP automatically sets this upon creation.

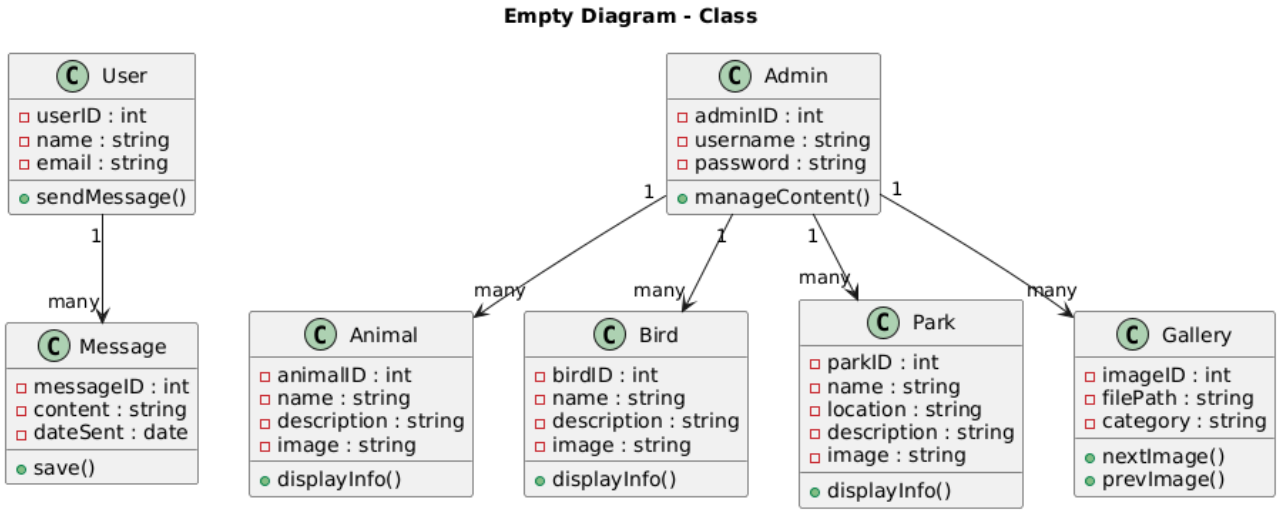
5.2 ER Diagram



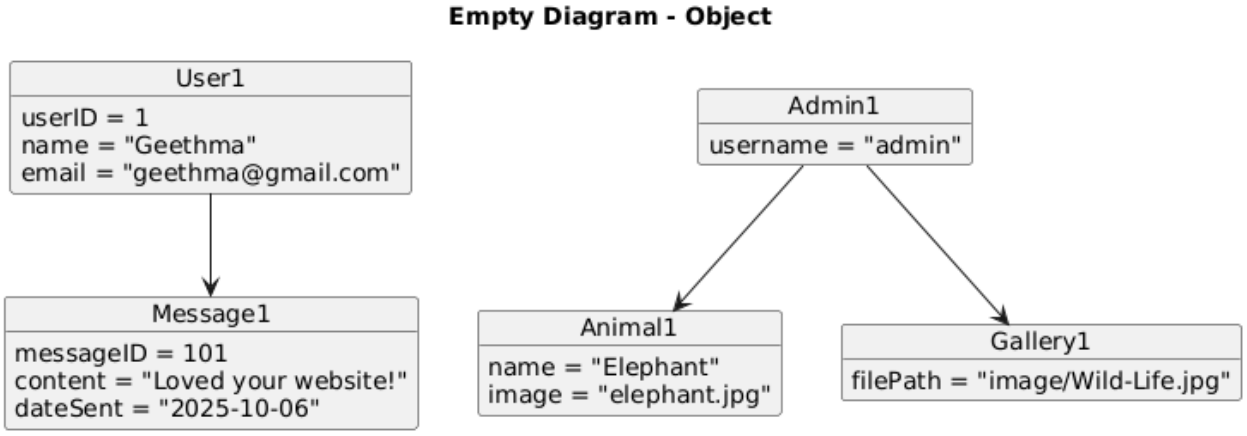
5.3 Use Case Diagram



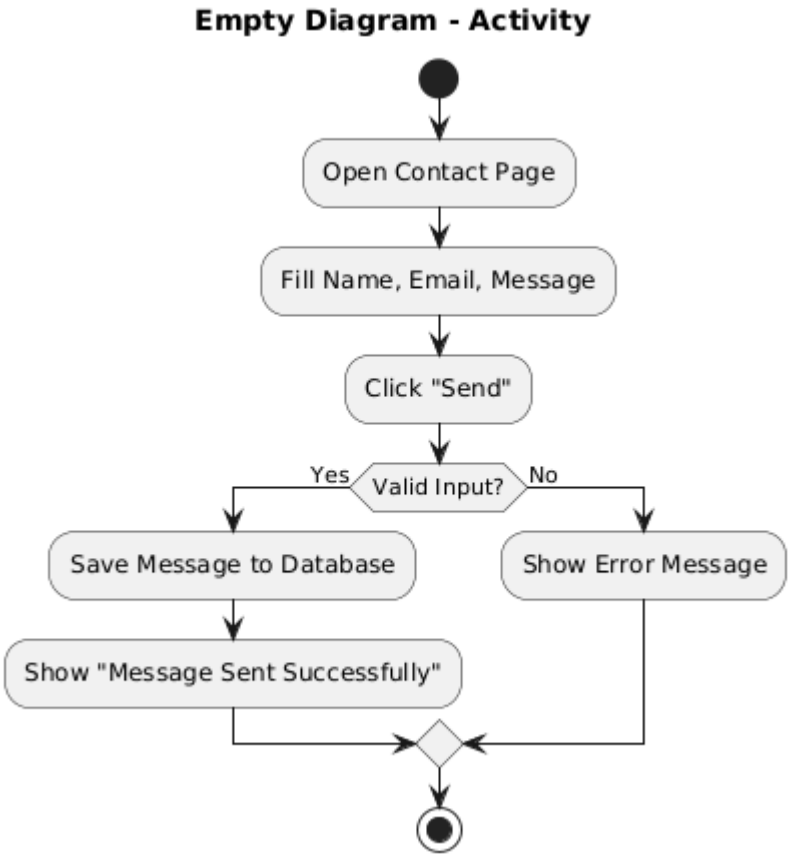
5.4 Class Diagram



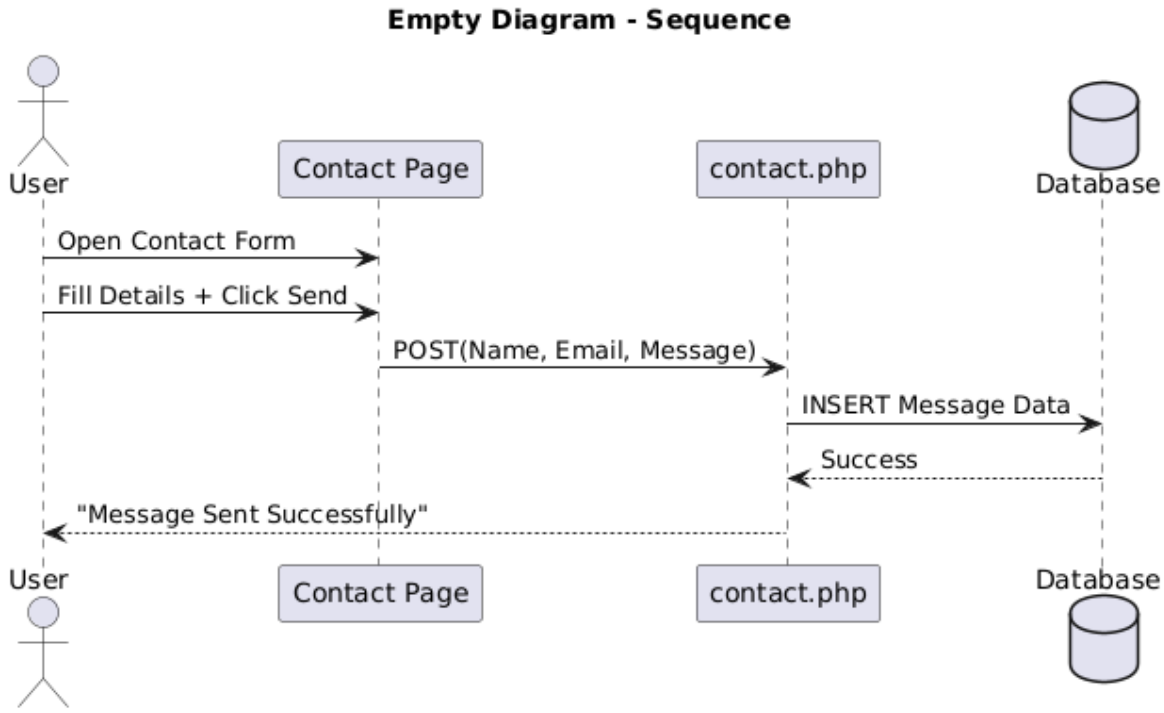
5.5 Object Diagram



5.6 Activity Diagram



5.7 Sequence Diagram



6. RESPONSIVE DESIGN

6.1 Mobile-First Approach

The website implements responsive design principles to ensure optimal viewing across devices:

- **Viewport Meta Tag:** Ensures proper scaling on mobile devices
- **Flexible Grid Layouts:** Cards adapt to screen size using CSS Grid and flexbox
- **Responsive Images:** Images scale proportionally to container width
- **Touch-Friendly Elements:** Buttons and interactive elements sized for mobile interaction

6.2 CSS Media Queries

Media queries adjust layouts for different screen sizes (desktop, tablet, mobile). Navigation adapts for smaller screens with potential hamburger menu implementation.

7. CHALLENGES AND SOLUTIONS

7.1 Challenge: Modal Functionality

Issue: Initial difficulty in dynamically populating modal content for different animals/birds/parks.

Solution: Implemented data objects in JavaScript containing all information, with functions that accept parameters to determine which data to display.

7.2 Challenge: Form Security

Issue: Concern about SQL injection and data validation.

Solution: Used prepared statements with parameter binding and implemented both client-side and server-side validation.

7.3 Challenge: Image Slider Smooth Transitions

Issue: Jerky transitions between images in the gallery.

Solution: Implemented circular array navigation using modulo arithmetic and added CSS transitions for smooth effects.

8. FUTURE ENHANCEMENTS

- **User Authentication:** Login system for registered users to save favorite animals/parks
- **Admin Dashboard:** Complete CRUD interface for managing content and messages
- **Search Functionality:** Allow users to search for specific animals, birds, or parks
- **Interactive Maps:** Google Maps integration showing park locations
- **User Reviews:** Allow visitors to share their wildlife experiences
- **Multilingual Support:** Full Sinhala and Tamil translations
- **Social Media Integration:** Share buttons for wildlife content
- **Newsletter Subscription:** Email updates about conservation efforts

9. CONCLUSION

The Sri Lankan Wildlife website successfully fulfills all assignment requirements by implementing a comprehensive web application with 6 pages, multiple JavaScript features, backend integration, and database connectivity. The project showcases Sri Lanka's rich biodiversity while demonstrating practical web development skills including HTML5 semantic markup, responsive CSS design, JavaScript interactivity, PHP backend processing, and MySQL database operations.

The website serves as an educational resource for wildlife enthusiasts and tourists while promoting conservation awareness. Through interactive features like modal popups, image sliders, and contact forms, users can engage meaningfully with the content. The technical implementation demonstrates proficiency in full-stack web development, from frontend user experience to backend data management.

This project lays a strong foundation for future enhancements and can be expanded with additional features such as user authentication, comprehensive CRUD operations, and advanced search capabilities. The modular code structure ensures maintainability and scalability for ongoing development.

10. REFERENCES

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